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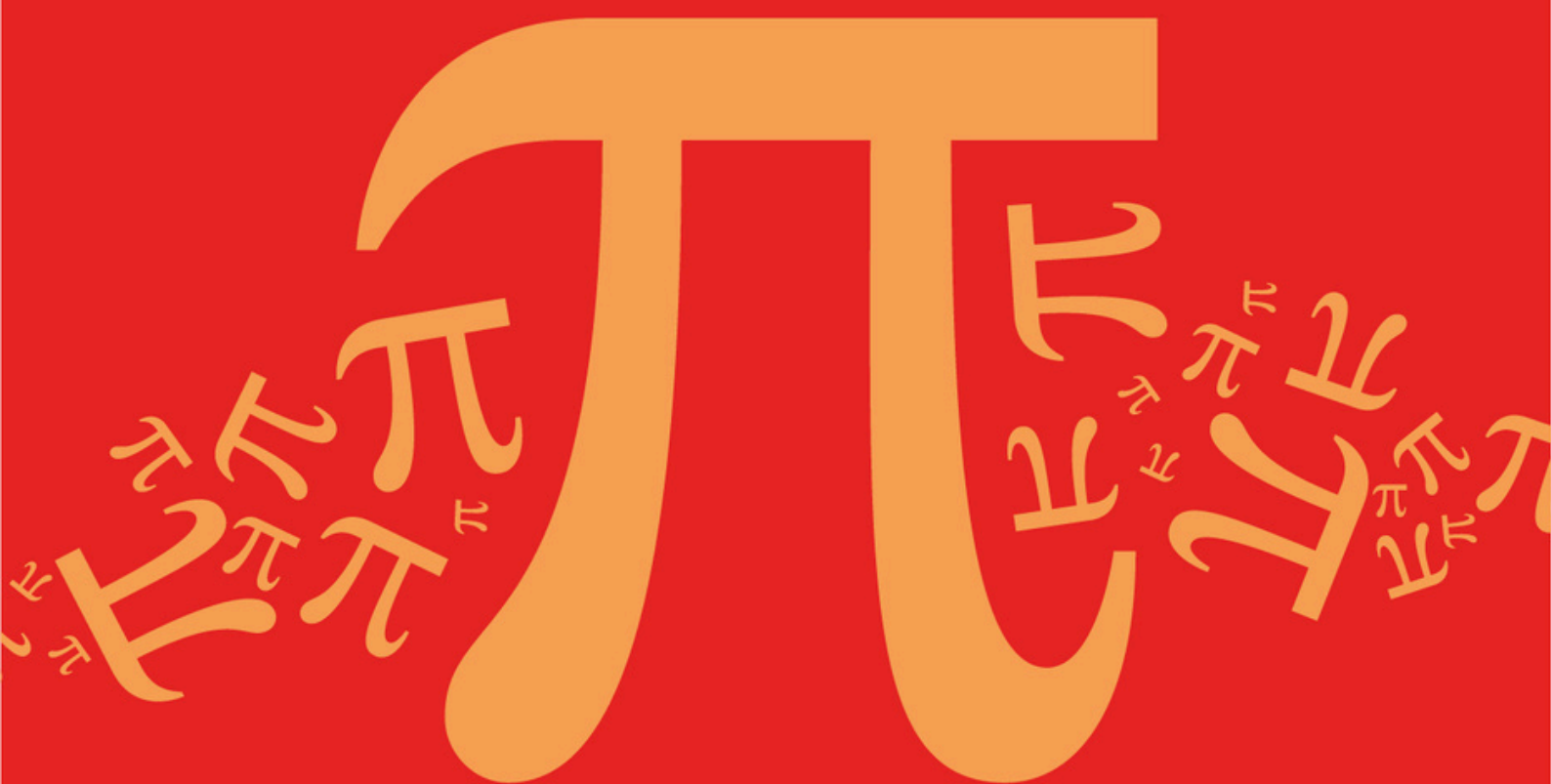
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Detailed mark scheme

Suitable for all boards

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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B6: Multiplication and Division in Modulus-Argument Form

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

B: Complex Numbers

Sub topic

B6: Multiplication and Division in Modulus-Argument Form

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Given that $z_1 = 4e^{i\frac{\pi}{3}}$ and $z_2 = 2e^{i\frac{\pi}{4}}$

state the value of $\arg\left(\frac{z_1}{z_2}\right)$

Circle your answer.

$\frac{\pi}{12}$
 $\frac{4}{3}$
 $\frac{7\pi}{12}$
 2

(Total 1 mark)

Q2.

(a) (i) Express each of the numbers $1 + \sqrt{3}i$ and $1 - i$ in the form $re^{i\theta}$, where $r > 0$.

(3)

(ii) Hence express

$$(1 + \sqrt{3}i)^8(1 - i)^5$$

in the form $re^{i\theta}$, where $r > 0$.

(3)

(b) Solve the equation

$$z^3 = (1 + \sqrt{3}i)^8(1 - i)^5$$

giving your answers in the form $a\sqrt{2}e^{i\theta}$, where a is a positive integer and $-\pi < \theta \leq \pi$.

(4)

(Total 10 marks)

Q3.

The complex numbers z_1 and z_2 are given by

$$z_1 = \frac{1+i}{1-i} \quad \text{and} \quad z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(a) Show that $z_1 = i$.

(2)

(b) Show that $|z_1| = |z_2|$.

- (2)
- (c) Express both z_1 and z_2 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. (3)
- (d) Draw an Argand diagram to show the points representing z_1 , z_2 and $z_1 + z_2$. (2)
- (e) Use your Argand diagram to show that

$$\tan \frac{5}{12} \pi = 2 + \sqrt{3}$$

(3)
(Total 12 marks)



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